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Case study on risk management practice in large offshore-outsourced Agile software projects

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Abstract: Information systems, offshoring and outsourcing have resulted in significant changes to software engineering practices. In this study, the authors explore industry practices in risk management in a large offshore-outsourced Agile Scrum software development project. The project consisted of distributed teams from two different IT vendors. Agile method is suitable for small and colocated teams, where the team members have high level of knowledge and commitment. In this study, the authors explore the challenges to Agile culture and practices posed by offshore-outsourced business model (where project ownership is shared, and business knowledge of the development team is limited), distributed project teams and large team size characteristic of offshore projects. In this study, it was further examined how these challenges were managed; and how the best practices can be used to improved performance in the context of offshore-outsourced Agile projects in general.

1 Introduction

1.1 Background of the study

Software development risk can be defined as an uncertain event or condition that has a negative effect on project outcome [1, 2]. As a result of globalisation, managing simple in-house projects have given way to new business models such as offshoring and outsourcing, where multiple IT teams and organisations work across geographies and cultures [3]. However, the new business models exacerbate existing risks and create new challenges [4, 5]. Therefore software risk management practices in IS offshoring and outsourcing is an area that receives much attention from researchers today.

Agile methodology is suitable for software development projects having small colocated teams [6, 7] that are in-house or insourced [8]. Each team member is expected to exhibit a high level of knowledge and project ownership [6]. The project employs a flat organisational structure [6]. However, the traditional offshore-outsourced context is different. Usually the offshore project teams have large team size, since the top reason for offshoring is the availability of low-cost skilled labour [8]. Although the offshore team's technical skills are usually good, their business knowledge and the understanding of project objectives are limited [3]. Multiple vendors may be engaged in the execution of a large project. The customer team and vendor teams are geographically distributed. In short, offshore-outsourced projects have a unique risk profile arising from challenges such as limited business knowledge and project ownership, larger team size, distributed teams,

as well as hierarchical command and control structure necessitated by customer–vendor relationship.

As reported in the literature survey in Section 1.2, there are many studies on challenges to Agile culture & practices, arising out of different aspects such as paradigm shift from the traditional software development method to Agile method, distribution of teams across geographies and offshoring under customer supervision. Today, IT offshore outsourcing is a business model that is steadily gaining momentum [9, 10]. In this model, a project is largely executed under the supervision of vendor [8]. Therefore we initiated a case study to profile the risks arising from offshore outsourcing and understand the corresponding risk resolution techniques adopted. We also analysed the risks and resolutions from the general perspective of distributed project management framework [11]. Our findings indicate that Agile projects are good candidates for the offshore-outsourced model, if risks unique to such a model can be identified, assessed and managed.

1.2 Related work

The works [12–14] provide an overview of Agile software development culture and practices. A detailed list of software risks involved in Agile projects is provided in the works [6, 7]. The case study [15] describes coordination requirements in large Agile projects. A hybrid model for Agile scrum projects is proposed in [16]. The study [17] investigates the critical changes needed for transitioning from traditional development model to Agile and enlists the following – change from heavily process-centric to short,

Table 1 Checklist of Agile Scrum risk factors in the offshore-outsourced context

Risk	Risk item	Refer	Traditional characteristics	Agile characteristics
1	control	[6, 15, 24]	process centric	people centric
2	management style	[6, 19]	command & control	leadership and collaboration
3	knowledge management	[6, 15, 19]	explicit	Tacit
4	role assignment	[6, 7, 15, 19]	individual – favours specialisation	self-organising teams – encourages role interchangeability
5	communication	[6, 15, 19, 24, 20]	formal and only when necessary	self-organising
6	development model	[6, 16]	lifecycle model (waterfall, spiral or their variations)	evolutionary delivery model (iterative and incremental)
7	desired organisational structure	[6, 20]	mechanistic (Bureaucratic with high formalisation)	organic (flexible and participative, encouraging cooperative social action)
8	team location	[6, 7]	predominantly distributed	predominantly collocated
9	traditional engineering measurements	[7]	project milestones (MIL-STD-1521), earned value	requirements burnt down/story completion
10	team size	[6, 15]	often >10	5–15
16	documentation	[6, 7, 15, 17]	substantial	less
12	resource utilisation	[7]	optimal (well-planned resource loading)	need continuous planning
13	customer involvement	[6, 19]	important usually during requirement analysis	critical and continuous
14	HR policies & processes – Rewards & Recognition	[6, 7]	suitable for projectised or matrix organisation [2]	reward system need to recognise performance; emphasise team performance over individual; and foster Agile culture
15	roles, responsibilities & skills	[6, 7, 15, 18, 19]	favours individualism	favours agile team culture
16	cost estimation	[7]	requirements are defined upfront. Therefore accurate estimates available	guess estimate in each sprint
17	quality control	[7, 17, 19]	most of the quality control activities are scheduled at the end of each phase	continuous quality control
18	stakeholder sign-off	[7]	required prior to starting the activities in a phase. Most important at the project start	not defined
19	contractual issues	[7, 16]	estimates are clear at the project startup, milestones are defined, and interim progress is measured in terms of earned value	estimates evolve and progress is not measurable in terms of earned value
20	turnover of key personnel	[8] interview finding	challenge	very high challenge

iterative, test-driven and people-centric development, changes in personal characteristics, changes in customer attitude, and changes in knowledge and education of stakeholders. See [18] for a comparison of Agile methods with the traditional methods with respect to interpersonal skills, business knowledge, team building and fostering of Agile culture. The case study [19] describes the effective use of tools and techniques for the collaboration of two geographically distributed teams. The case study [20] describes how to build and sustain Agile offshore teams. Based on the above studies, we compiled a checklist of top risks that may be applicable to ‘large Agile offshore-outsourced’ projects. We used this checklist as a tool for our interviews (see Table 1).

From a systematic survey of 72 rated journals, Persson *et al.* [11] developed an integrative framework for managing risks in geographically distributed software projects (GDSP). The framework has two dimensions – risks and risk resolution techniques. Persson *et al.* identified eight risk factors – task distribution, knowledge management, geographical distribution, collaboration structure, cultural distribution, stakeholder relations, communication infrastructure and technology setup. Applying McFarlan’s [21] concepts, Persson *et al.* divided the risk resolution techniques into four categories – planning, control, social integration and technical integration. We used this framework in our case study to explain the risk resolution techniques observed from a broader GDSP perspective (see Tables 2–5).

2 Research design

2.1 Research method

Prior studies on software development risks have used methods such as Survey of IT professionals, Delphi study, Case studies and Action research. Some researchers have used personal experience in the industry or secondary sources of information. We conducted a single embedded case study. The characteristics of the project selected are shown in Table 6. The project comprised four teams – the customer team, two teams belonging to one vendor and one more team belonging to a different vendor. The two teams belonging to one vendor (see Table 6, Team B and Team C) formed the case. The units of observation were risk factors and risk resolution techniques, from the perspective of offshore development centre (ODC) team. The study was designed according to the procedure laid out in [22].

2.2 Characteristics of the project selected for case study

A large, Agile scrum, offshored-outsourced software project was selected for the study. The numbers and description about the project provided below are indicative in nature. They have been modified, preserving the original characteristics, in order to ensure confidentiality of project, customer and the project team. The project under

Table 2 Risk resolution techniques: planning

R#	Item	Description	Case study finding	More details
P11	acquire complementary skills	highly developed collaboration skills	the team had basic capability in this area. The junior team members were mentored. Scrum masters & onsite coordinator acted as bridge builders	
P21	adjust meetings to distributed context	identify subjects of the meeting subsequently to determine the relevant participants	all meetings were agenda driven. Only relevant persons whose opinion counted in the context of meeting, were called for meetings	
P22	adjust meetings to distributed context	determine the requirements of the meeting. If visual aids are necessary, a videoconference may be appropriate	audio or video conferences were used for the sprint planning/sprint review meetings. Audio conferences were used for project management reviews	
P22	adjust meetings to distributed context	ensure well-prepared participants by sending agenda and other important documents prior to the meeting. Participants expected to have material ready for the meeting	the agenda and material for review were circulated beforehand. The participants came prepared for the meeting	
P23	adjust meetings to distributed context	elect a presiding officer (PO) of the meeting to ensure appropriate turn taking and compliance of the agenda	the meetings were presided over by persons, based on their roles. But compliance to the agenda and turn taking were ensured	
P24	adjust meetings to distributed context	the PO should summarise discussions periodically during meetings	periodic summarisation was done for clarity, convergence and steering of the meeting	
P25	adjust meetings to distributed context	in the case of a telephone conference: the PO should describe activities in the room, people coming and going, and drawn objects	the name of participants were announced automatically by the teleconferencing system	
P26	adjust meetings to distributed context	alleviate drawbacks, that is, in terms of odd work hours, of time zone differences by using fixed meeting schedules	cross-site meetings were pre-scheduled. Changes in the schedule/agenda, if any, were announced in advance	
P31	divide tasks systematically between sites	focus on initial analysis of the modular structure to identify dependencies and predict the effects of changes	1. at project startup, overall requirements were defined and high-level design prepared (system architecture, database & interfacing needs)	4.1. Hybrid development model
P32	divide tasks systematically between sites	base the task distribution on a well-devised modular design that takes the project structure into account. Segmented tasks should be well understood and stable	2. a POC program was done to test the design	
P33	divide tasks systematically between sites	define architecture 'light' with few, but important principles. The architecture should describe the entire system on a high level	3. work distribution aimed at ensuring maximum cohesiveness of modules within site and minimal coupling across sites	
P41	reduce coupling between sites	reduce dependencies between sites – (a) provide well-defined work areas and thereby give the participants minimal control, or (b) give extensive responsibility and full control	the work areas were well-defined, and sites were loosely coupled	4.1. Hybrid development model
P51	create shared collaboration platform	utilise a shared tool to track bugs and corrections in every part of the distributed organisation, thereby providing transparency	common tools for defect tracking, and issue tracking across the sites	
P61	establish shared goals	the PM should establish and obligate the participants to a shared project goal to develop a common identity	periodic project management review meetings were conducted	
P62	establish shared goals	the PM should act both proactively and creatively by periodically updating participants about progress towards planned goals	a project management tool 'Rally' was used to capture and display the project status & progress. Bulletins on status & progress were issued periodically. Monthly project management reviews were done	
P71	establish communication norms	create formal guidelines for the use of asynchronous media describing the communication content and how it should be performed	guidelines, formats and templates were maintained. The documents were subjected to reviews. A version control system was in place	
P81	define roles and responsibilities	at project startup, define undisputed areas of responsibility for all participants as well as the relational roles being instituted	at project startup, the organisational structure, interfaces and roles & responsibilities were defined	
P82	define roles and responsibilities	the PM should exercise authority to ensure the completion of delegated assignments	progress and quality were monitored through project status reports from 'Rally' tool, and project management reviews. PM had the final word with respect to assignments/delivery	
P91	reduce time zone differences	–	the locations were pre-determined ODC sites. Work distribution was done with a view to reduce the level of cross-site interactions. A three hour overlap was maintained for coordination between offshore and customer site	

Table 3 Risk resolution techniques: control

R#	Item	Description	Case study finding	More details
C11	focus on deliverables	plan and control critical deliverances perfectly considering differences in, e.g. tools and data format	a delivery checklist was in place. Reports on physical configuration/ functional configuration audits were submitted along with the delivery	
C21	establish task communication & control between sites	adjust the project structure dynamically to the individual phases of the project to avoid bottlenecks and waste of resources	the inter-site interaction requirement was limited. Scrum masters and onsite coordinator acted as bridge builders for the development team	4.4. Communication & control
C31	maintain site autonomy	make sites self-managing by establishing a system that allows participants to monitor own processes	burn down charts, project velocity graphs, and daily/weekly meetings provided visibility to the team's progress/issues and thereby control over them	
C32	maintain site autonomy	introduce team empowerment	team members actively participated & contributed to daily standup meetings. They selected their task & proposed estimates. Innovations were fostered; rotation of roles was done	4.3. Self-organising team
C41	establish shared control mechanisms	design a method to monitor and control the information flow concerning project status, providing information to all stakeholders including individual groups	a well-defined project communication plan was in place. This included, stake holder roles & responsibilities, reporting needs, frequency of reporting, and escalation procedure, among others	4.4. Communication & control – 'agile specific'
C42	establish shared control mechanisms	let participants themselves monitor changes in their environment	build tracking and daily progress tracking were done. These roles were rotated among the team members	4.3. Self- organising team
C43	establish shared control mechanisms	the PM should make sure that periods of inactivity are detected and reported automatically	daily time sheets, daily standup meetings and resource availability chart (team, reviewers and subject matter experts) helped to detect idle time. 'Look ahead planning' of work further helped in this	4.2. Resource utilisation
C44	establish shared control mechanisms	deploy tools to measure trust in the project's early stages	–	
C45	establish shared control mechanisms	emphasise focus on results and performance compared to traditional projects	individual and team performance were transparent. Techniques such as post it notes on accomplishments; facilitation in weekly meetings; project velocity/ burn down charts emphasised result orientation	4.5. Rewards & recognition
C51	establish temporal communication & control mechanisms	focus on synchronisation, plans, and procedures in the development process, enabling transferring of tasks from one site to another	CMMI processes and UML techniques helped seamless task synchronisation. The inter-site interaction requirement was limited. Therefore asynchronous communication was sufficient	4.6. Process discipline
C61	maintain project organisation overview	a clear plan of communication based on individual and group communication needs. The plan should further contain a list of contacts and their professional and decision making competencies	see C41	
C71	maintain task overview within and across sites	define architecture 'light' with few, but important principles. The architecture should describe the entire system on a high level of abstraction	at project startup, overall requirements were defined. A high-level design was prepared, including system architecture, database design and interfacing needs	4.1. Hybrid development model
C72	maintain task overview within and across sites	at project startup: Define clearly, which groups are involved, and which tasks they work on	the teams had well defined scope of work (as a result of due diligence given to work distribution, at project startup and subsequently during execution)	4.1. Hybrid development model
C73	maintain task overview within and across sites	create a project website that summarises project content, progress, planning and group-related information		
C74	maintain task overview within and across sites	use a content management system	share point was used for communication & collaboration	4.6. Process discipline
C75	maintain task overview within and across sites	evaluate the task complexity considering task structure, task environment, and internal and external coupling	done accordingly	
C81	monitor and improve communication	–		

Continued

Table 3 Continued

R#	Item	Description	Case study finding	More details
C91	maintain a supportive environment	use a supportive non-dictating management style	the management was responsive. participative culture was fostered	4.3. Self-organising team
CA1	analyse and manage errors	causal analysis	defects were collected and analysed periodically. Accordingly, corrective actions were taken to improve project performance and product quality	

Table 4 Risk resolution techniques: social integration

R#	Item	Description	Case study finding	More details
S11	improve capability to manage cultural differences	adjust management style according to culture, e.g. participants' preferences for well-defined tasks versus loosely defined task and self-management	tasks were defined. During the last week of a sprint in progress, 'pre-planning meetings' were organised to discuss & refine user stories scheduled for the next sprint. The scrum master provided mentoring the team had basic capability in this area. The junior team members were mentored. Scrum masters & onsite coordinator acted as bridge builders the ODC had basic capability in this area. In addition, scrum masters and onsite coordinator acted as bridge builders connecting the offshore team with other sites and customer team	4.4. Communication & control
S21	improve distributed collaboration skills	–	several Programs were initiated by HR for team building	4.4. Communication & control
S31	improve language skills	–	at project startup, video conferences were used for team introductions & project induction. Periodic video conferences were held by the technical team, which also contributed to relationship building	4.8. Imbibing Agile culture
S41	emphasise early teambuilding activities	stimulate the interaction between participants from project startup	–	
S42	emphasise early teambuilding activities	arrange videoconferences if face-to-face meetings are impossible	–	
S43	emphasise early teambuilding activities	create cross functional groups in the initial phase of the project to encourage social relations across areas of expertise	–	
S51	promote humour and openness	stimulate relationship building by using humour	weekly entertainment programs ('Friday fun at work') and periodic day-out programs were organised by HR for party games and social jelling see S51. This helped to improve openness	
S52	promote humour and openness	promote openness to avoid hidden agendas that impede productivity	the new team members and junior team members were given a chance to speak outside of the daily standup meetings. Mentors were assigned and they hand held them for an initial period of one sprint to build up competency and confidence	
S61	use mentors to integrate new members	use mentors to integrate new participants. The mentor is responsible for social adaptation and communication of the project and group's history and values	–	
S71	use face-to-face meetings appropriately	prioritise face-to-face meetings to develop trust and shared identity easier and faster	–	
S71	use face-to-face meetings appropriately	the PM should travel to all distributed sites and keep closest to the sites where task coupling is high and group cohesion low	the customer PM visited the sites in person two times during the project execution. But it may be noted that inter-site task coupling was minimal	
S81	develop liaisons between sites	include travel expenses in the overall budget and do not perceive them as additional costs	travel expenses were anticipated and built into the estimates at project startup. Prior outsourcing experience helped in this inclusion	
S91	adopt shared reward systems	–	–	

consideration was part of a 25 million USD initiative for the consolidation of business applications of a large healthcare provider in the USA. The project exhibited a broad range of features relevant to the study. The duration was 18 months. The project was outsourced to two different vendors. The

vendors were certified at CMMI level 5 – indicating the highest level of process capability and maturity [23]. This was done to minimise the influence of random factors related to the capability of software development processes. The team size was large, with 26 members. There were four

Table 5 Risk resolution techniques: technical integration

R#	Item	Description	Case study finding	More details
T11	increase technical compatibility between sites	–	–	
T21	standardise and train in methods across sites	establish technical standards and stick to them to maintain project integrity	common development standards and guidelines were developed and used across sites	
T22	standardise and train in methods across sites	introduce shared guidelines for error handling, accessibility to other group's documentation, and documentation for tests and testability	common guidelines were used for error handling. The teams used common templates for test specification & test results. The documents were shared using share point, with different access rights defined for different vendor teams	
T22	standardise and train in methods across sites	use a shared tool that allows for tracking of corrections and bugs in all parts of the distributed organisation	common tool used for defect tracking	
T23	standardise and train in methods across sites	standardise tools, methods and processes to create a harmonic and efficient project organisation	CMMI level 5 processes were used. Common tools were used for defect tracking, issue tracking, document sharing, and configuration management, across the sites	
T31	adopt appropriate communication technologies	choose ICT that supports a suitable atmosphere for creation of trust, e.g., videoconferences can reduce the sense of physical and psychological distance	video conferences were used for induction; for review meetings, and to discuss major issues. (It may be noted that between site dependency was less.)	
T32	adopt appropriate communication technologies	support creativity by choosing ICT that supports a decentralised communication network, e.g., instant messaging	instant messaging, chat and email were extensively used, reducing the volume of formal documentation	
T33	adopt appropriate communication technologies	utilise videoconference or telephone conference in addition to group support systems usage, whenever the group is working on a decision making task	video conferences were arranged to discuss major technical challenges as well as periodic project management reviews	
T34	adopt appropriate communication technologies	use telephone- and videoconferences to support relationship building	video conferences were used for initial team introductions – teams, stake holders, supporting groups. It may be noted that between site dependency was less	
T35	adopt appropriate communication technologies	choose lean media, such as e-mail, for simple messages to reduce noise in the information. For complex and debatable information, choose richer media	emails, instant messages, and chat were used extensively for informal/ personalised/brief communication; Formal documents were shared using SharePoint	
T36	adopt appropriate communication technologies	larger projects (in terms of number of participants) should adopt communication technology that supports communication & control and logistics, whereas smaller projects should choose technology that supports collaboration and communication	it may be noted that cross-site dependency was less. Share point was used for formal communication & collaboration, especially across sites. Informal communication extensively used within a vendor team	
T41	improve collaboration and communication technology skills	–	the junior team members were obtained mentoring from scrum masters	
T51	improve development technology skills	–	periodic trainings were organised, by vendor teams for their team members. Corporate competency development team also provided support and organised training	
T61	handle differences in methods between sites	communicate knowledge of methods and processes deployed by other groups and enhance understanding of strengths and weaknesses	best practices, and lessons learned were shared across teams	
T62	handle differences in methods between sites	'capability maturity model' (CMMI) can be used to ensure equal quality level of the deferring methods	CMMI processes were used by both vendors	4.6. Process discipline
T71	combine waterfall model and prototyping	combine the waterfall model and prototyping, using prototyping to determine requirements and the waterfall model to maintain modularity, low task coupling, and structure in the process	hybrid development model was deployed. Sequential waterfall processes included high-level design at project start up, and final tests during the delivery of each increment. Detailed design and development was done iteratively in sprints	4.1. Hybrid development model

Table 6 Project demographics

Team	Who	Location	Team-size (offshore)	Team-size (onsite)	Responsibilities	Remarks
A	customer	USA		4	• overall project ownership • high-level architecture • proof of concept • final tests: integration & validation	team details:- 01 product owner (PM) 01 business Analyst 02 subject matter experts (the ODC teams provided support for final tests) team details:-
B	vendor-1	Chennai, India	7			
C	vendor-1	Chennai, India	7	1	• detailed design • development • support to final tests	offshore PM (not billed) + 01 onsite coordinator + 02 scrum masters + 12 offshore team members team details:-
D	vendor-2	Bangalore, India	6	1	• detailed design • development • support to final tests	01 onsite coordinator 01 scrum master 05 team members

teams. Team-B and Team-C, working under Vendor-1, were selected for this study. See Table 6 for project demographics.

The project processes included the following:-

1. 'Well-defined "overall" project scope';
2. 'Sound architecture and key framework components keeping in mind the bigger picture' that is, the overall scope and work distribution to multiple teams;
3. Joint Analysis & Design (JAD) sessions were organised to capture high-level requirements for initial estimation and subsequently on demand;
4. 'Adaption of changes was done in a flexible manner';
5. 'Efforts were taken to maintain close collaboration among team members';
6. Business users kept close interaction with developers to avoid gaps and errors;
7. 'A culture that supported negotiation, trust in people and collaborative work environment was fostered'.

The tools used in the project for communication and collaboration included (but not limited to) the following – Rally for application lifecycle management for distributed Agile development; Star Team for software change and configuration management; SharePoint as a centralised repository of documents, for sharing across the project; and WebEx for live meetings and demonstrations across the distributed teams.

2.3 Preparation of risk checklist for conducting the interviews

The project under consideration consisted of three geographical locations. However, it may be noted that inter-site coupling was minimal, and the vendor teams worked with a high degree of independence. Therefore we gave more focus to the influence of offshore-outsourced context and large team size, on Agile culture and practices. We prepared a draft check list of risks applicable to Agile projects in the offshore-outsourced context, from the works discussed in Section 1.2, viz. [6, 7, 15–20, 24]. The checklist was reviewed by 'senior IT professionals'. We incorporated their feedback to arrive at the final checklist as is given in Table 1. See Section 4 for further description of the risks in the project context.

2.4 Interview protocol

The first author, leveraging his network, obtained senior management approval for conducting the case study. An email stating the objectives of the study and providing assurance for maintaining confidentiality of information was sent to the management. First degree data [22] were collected from interviews. Third degree data [22] were obtained from documents & records such as statement of work, project plan, risk management plan, minutes of sprint planning meetings/sprint review meetings, burn down chart, project velocity, resource utilisation reports, project overview presentations and project closure reports. The case study used formal template-based approach for data analysis, which is best suited for software engineering case studies [22]. To demonstrate evidence for conformance to qualitative data analysis, the following are provided – citations from interviewees, figures indicating real project environment, project and process metrics, as well as data classification and summarisation from observation to findings with respect to the units of study, viz., risk factors and risk resolution techniques. In order to ensure confidentiality, changes were made to the numbers and description, while preserving the original characteristics. Original records of interview were destroyed after finalising the report.

The ODC PM and two scrum masters were selected for the interview. The risk checklist shown in Table 1 acted as a guideline to maintain structure and focus. At the same time, flexibility of explanation was maintained. The first author and two senior technical architects with more than 12 years of IT experience together constituted the interview team. The team had long-term association with similar IT projects. Confidentiality of information was assured to the interviewees through an email. First phase of interviews were done in person taking 90–120 min. Opening the interview session, the topic was introduced and assurance on confidentiality of information gathered was provided. A pyramid model was used for discussions [22]. The interviewee was taken through each item in the checklist. Iterative and in-depth investigation of observations was carried out. The experiences ('lessons learned') and 'best practices' in the project were captured. Notes were taken manually. Audio/video recording was not done (since we did not obtain permission to do that). Relevant information from project records and project metrics were noted. After

Table 7 Profiling risks & resolutions, attributable to offshore outsourcing

Section no.	Section name	Risk	Risk item	Agile characteristics	Case study finding
2.2	characteristics of the project selected hybrid development model	10	team size	5–15	26 members divided into four teams hybrid model (linear, incremental & iterative): sequential processes were planned at the beginning and end of the project with agile model sandwiched in between at project startup, overall business functionality was studied and high-level estimation done. Subsequently, sprint wise estimation was done acceptance criteria for delivery was predefined
4.1		6	development model	evolutionary delivery model (iterative and incremental)	
4.1	hybrid development model	16	cost estimation	guess estimate in each sprint	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.1	hybrid development model	18	stakeholder sign-off requirements	not defined	
4.2	resource utilisation	12	resource utilisation	need continuous planning	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.3	self-organising team	2	management style	leadership and collaboration	
4.3	self-organising team	4	role assignment	self-organising teams encourages role interchangeability	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.3	self-organising team	15	roles, responsibilities & skills	favouring Agile team culture	
4.3 & 4.6	process discipline	1	control	people centric	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.4	communication & control	5	communication	self-organising	
4.4	communication & control	7	desired organisational structure	organic (flexible and participative, encouraging cooperative social action)	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.4	communication & control	8	team location	predominantly collocated	
4.5	rewards & recognition	14	HR policies & processes	need to recognise performance; emphasise team performance over individual; and foster Agile culture	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.6	process discipline	3	knowledge management	Tacit	
4.6	process discipline	11	documentation	minimal	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.6	turnover of key personnel	20	challenge	very high challenge	
4.7	quality control	13	customer involvement	critical and continuous	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos
4.7	quality control	17	quality control	continuous quality control	
4.9	contract type	9	traditional engineering measurements to measure project progress	use of requirements burnt down/story completion, to measure project progress	‘planned look-ahead meetings’ to determine resource requirements ‘pre-planning meetings’ to refine user stories and identify impediments collaboration & leadership suitable for agile culture, blended with command & control needed for coordination in a large & distributed project limited rotation of roles. The scrum master role was not rotated; acted as facilitating team lead. Some other roles rotated availability of competent team members, who can play multiple roles, was a challenge people centric and also process centric horizontal & vertical coordination requirements were identified and plans made accordingly hybrid organisational structure amalgamating flat structure for agile culture with hierarchical structure needed for coordination large/distributed project teams distributed in three locations. The teams were loosely coupled attempts made to improve the feedback process, and to modify the reward system to foster Agile culture/transparency. Individual performance was given primary importance, but team performance was also counted both informal and formal (use of CMMI, UML guidelines) level of documentation was medium employee release at regular intervals and employee turnover had influence on project performance. Process discipline was used as an effective mitigation strategy continuous collaboration measures for continuous quality control included –test-driven development, continuous integration, early review of completed user stories, allocation of a permanent testing specialist T&M contract was used, supported by requirements burnt down chart & sprint demos

Continued

Table 7 Continued

Section no.	Section name	Risk	Risk item	Agile characteristics	Case study finding
4.9	contract type	19	contractual issues	estimates evolve and progress is not measurable in terms of earned value	T&M contract was used High-level initial estimates were prepared. Sprint planning meetings were used for sprint wise estimation

the interview, we compiled the notes and exchanged them. We conducted an internal discussion and noted down points requiring further clarification. A follow-up discussion was done with the PM in person to obtain clarifications and more insights. Follow-up discussions with the other two interviewees were conducted using 'skype' facility, over internet. The summary of interviews was prepared after a second round of internal discussions and circulated to the interviewees through email. After obtaining the feedback from the interviewees (through email and phone), the data were analysed and a case study report was prepared. The report was shared with the interviewees and finalised based on their feedback.

2.5 Validity & limitations

The measures taken to improve validity included the following [22]. The interview checklist was prepared based on the literature survey on the research topic under consideration; and it was refined based on the feedback provided by 'Senior IT professionals'. This enhanced the construct validity. During data analysis the observations were mapped to detailed risk resolution framework for globally distributed software projects [11] for comprehensiveness of coverage. The interviewers were associated with similar IT projects for a prolonged time. Therefore the interviewee's observations were expected to be well understood. The case exhibited wide variations (see Section 2.2). Data triangulation was achieved through the following. To achieve observer triangulation, multiple team members were interviewed. Also, personnel with different roles were selected for the interview. To achieve source triangulation, notes were taken from the project records and documents, as well as interviews. To achieve methodological triangulation, qualitative and quantitative techniques were used. The qualitative techniques included interviews. Quantitative techniques included analysis of project metrics and project performance.

However, the study is constrained by the limitations that are usually associated with case studies [22]. Although a representative case showing wide variation of features was selected for the case study (see Section 2.2), it may be noted that there are a multitude of features associated with software risk management, all of which were not considered. While we could obtain the perspective of ODC team, we were not able to contact the customer team and obtain their perspective.

3 Case study findings

The case study observations were analysed to capture the risks and risk resolution techniques adopted in the project. Section 3.1 deals with challenges attributable to offshore outsourcing of Agile projects. Section 3.2 deals with the

challenges from a broader perspective of distributed project management.

3.1 Risk profile attributable to offshore outsourcing of Agile projects

Observations from the interviewees based on the risk checklist for Agile projects (see Table 1) were analysed and reclassified [22], as shown in Table 7. The risks, corresponding risk resolution techniques adopted in the project, and the subsection where detailed discussion can be found are shown in Table 7. See Section 4 for detailed discussions.

3.2 Risks & resolutions from distributed project management perspective

The project selected for the case study consisted of multiple geographical locations. We mapped the risks and risk resolution techniques observed in the project, to the integrative framework for managing risks in GDSP as proposed by Persson *et al.* [11]. The risk item, risk description and case study findings are shown in Tables 2–5. If innovations specific to 'Agile offshore outsourcing' were observed, the detailed discussion is provided in the subsection shown under the column 'More Details'. They are further discussed under Section 4.

4 Discussion of results

Agile projects imbibe the characteristics of iterative and incremental models. The offshore-outsourced context necessitated the inclusion of linear model as well, with a set of sequential activities at the beginning and end of the project (see Fig. 1 and Section 4.1). To summarise, the project imbibed characteristics of all the three basic software development models. Contrary to the prescriptions of Agile model, the project was large and not collocated. The IT vendors had a hierarchical organisational structure and are more aligned to traditional software development. The project team was distributed across geographies, although inter-team coupling was limited.

We mapped the risks and risk resolution techniques observed in the project, to the integrative framework for managing risks in GDSP as proposed by Persson *et al.* [11]. The risk item, risk description and case study findings are shown in Tables 2–5. Risk resolution techniques specific to Agile offshore-outsourcing context, if any, are marked under the column 'More Details'; and explained under Sections 4.1–4.9. The framework lists 66 risk resolution factors, categorised under the broad umbrella of planning, control, social integration and technical integration [21]. However, it may be noted that the inter-team coupling was minimal and the vendor teams worked with a high degree of independence. Therefore detailed treatment based on

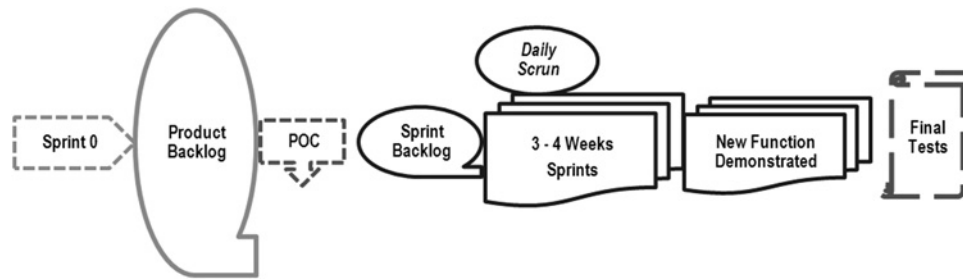


Fig. 1 Hybrid development model used in the project

individual risk factors proposed by Persson *et al.* [11], viz., task distribution, knowledge management, geographical distribution, collaboration structure, cultural distribution, stakeholder relations, communication infrastructure and technology setup, was not included in the scope of our work.

To understand the risk profile and risk resolution techniques unique to Agile practice adaption in the offshore-outsourced context, observations from interviewees on the 20 items risk checklist were analysed and reclassified as shown in Table 7. These risks and the corresponding risk resolution techniques adopted are discussed in detail under Sections 4.1–4.9.

4.1 Hybrid development model

In order to address the challenges arising from offshore-outsourced model, large team size and lack of collocation, some of the practices from traditional sequential development method were integrated in to the Agile method to form a hybrid model (see Fig. 1). Sequential processes were planned at the beginning and end of the project with Agile processes sandwiched in between. This model has certain similarities to the one proposed by Hayata and Han [16]. In Fig. 1, the components marked with ‘–’, represent the ‘gates’ that indicate traditional sequential processes. The gates included the following – an elaborate Sprint 0, Proof of Concept (POC) and Final Tests. The work executed at offshore is shown in blue outline. It may be noted that three teams worked simultaneously, each progressing with different sprints at a time.

The project started with a ‘detailed sprint zero phase’. The required business functionality was defined more elaborately than that is usual with Agile practice. The software architecture and interfaces were defined and database design was completed. A POC program was undertaken ‘to ascertain the feasibility’ of the design. The solution was refined from the ‘lessons learned’. The activities that could be outsourced were determined. High-level estimates and ‘software release plan prepared’. This helped to determine the work distribution to vendors and prepare a statement of work for outsourcing. Criteria for work distribution included minimisation of across site coupling; and maximisation of cohesiveness of modules within a site, with respect to software development. The acceptance criteria for delivery were defined at project startup. The software was delivered in three increments. Prior to the release of each increment, all the related sprints, from the two participating vendors were brought together for a set of Final Tests, which included, integration test, system test to measure technical performance and functional validation.

4.2 Resource utilisation

The following measures were taken to reduce idle time. Planned ‘look-ahead meetings’ were organised to set ‘focus on next release’ and thereby the resource requirements. During the last week of a sprint in progress, ‘pre-planning meetings’ were organised to discuss and refine user stories scheduled for the next sprint and identify/resolve impediments. These measures ensured predictable work load, minimal idle time and improved efficiency of the ODC team. Programs for competency development (in technical skills, soft skills and business knowledge) were planned to care of any idle time that remained unavoidable.

4.3 Self-organising team

The project had roles such as architect, scrum master, designer, developer, tester, business analyst, test specialist, onsite coordinator and certain facilitating roles. Some team members played multiple roles. According to [6, 19], the spirit of Agile methodology is best implemented in a team having a flat organisational structure. However, in the case under consideration, we observed a combination of flat and hierarchical structures. In the Agile method, the role assigned to a person is rotated on a regular basis, in order to improve the sense of ownership and to develop individual skills. In an ODC, resources are often allocated to a project from the organisation’s central ‘talent pool’, with an objective of achieving an optimal mix of team members with differing competencies. Thereby, the team as a whole is capable to deliver. However, ‘availability of competent team members who could play multiple roles’, remained a challenge. Hence, no scheduled rotation of roles was planned. The scrum master was a facilitating team lead and the role was not rotated. A business analyst cum test specialist was permanently allocated to the project. The team members were allowed to take up stories by themselves, ‘most of the time’. Developer and tester roles were interchanged based on practical needs. Roles such as build tracker, the daily progress tracker and moderator were rotated. ‘Achievers’ were given higher responsibilities.

4.4 Communication and control

The project was large, and multiple teams from multiple vendors were working across geographies. Therefore a hybrid organisational structure was deployed, blending flat structure suitable for Agile culture, and hierarchical structure needed for coordination in a ‘large’ and ‘offshore-outsourced’ project [15]. The team members reported to the scrum masters and the scrum masters reported to the ODC PM. In order to achieve horizontal coordination, team building exercises were planned, which

is discussed in detail under Section 4.8. In order to achieve vertical coordination, the following measures were taken. A 3-h overlap was maintained in the working time between offshore and onsite, to facilitate mutual communication. The product owner worked as a boundary object or 'bridge builder' [15] between the user and development team [24], and coordinated the distributed teams. The offshore PM acted as facilitator to interface the teams under the same vendor, the teams and internal support groups in the vendor organisation, as well as the ODC and customer accounts office. An onsite coordinator acted as a bridge builder between offshore team and customer team.

4.5 Rewards and recognition

Implementing Agile-compliant performance evaluation in traditional organisation structure was a challenge. The ODC had a reward and recognition system based on personal performance. Therefore contrary to the Agile practice, individual performance was given more importance. But team performance was also given weightage. Attempts to align the system with Agile culture included, but not limited to, the following measures – basing the competency requirements on band (cadre or position in the organisational hierarchy), defining supplementary skill requirements for different roles, and recognition for guidance & mentoring. Within the project, attempts were made to achieve transparency in performance recognition through measures such as the following – public felicitation of performers by posting appreciation notes on the seating location of the employee (before the person arrives for work), appreciation in monthly team meetings, award of gift vouchers and nomination of performers for training programs. Subjectivity existed in estimating the user stories, accommodating 'the development complexity' (which differed from one set of modules to another) and measuring individual productivity. To quote one of the interviewees, 'those people who made more noise got more rewarded'. One of the measures taken to reduce subjective bias was 360-degree feedback [25], which included collecting feedback from other sources, viz., peers, personnel at lower rungs of hierarchy and corporate support groups. 'Skip a level meeting' was organised periodically, where the team meets with the management at a level higher than the immediate supervisor.

4.6 Process discipline

According to Turner and Boehm [7], Agile methods are light weight processes, and formal documentation would make them less effective. In Agile practice, tacit knowledge and collaboration are important in knowledge sharing. In the context of offshore-outsourced projects, the team was not collocated. Therefore additional measures were required for collaboration among teams and with the user. Retaining knowledge within the Agile team for longer duration was challenging because of the transient nature of resource allocation. Employees were allocated to a project for a definite period of time and they were released, based on vendor's organisational policy/requirements. Employee turnover, reportedly at about '15% per year', was another challenge. The team was expected to get disbanded after final software delivery, leaving a knowledge gap, when transitioned to maintenance phase. Some of the risk mitigation strategies devised to address the knowledge gap arising from transient nature of employee allocation were:-

1. Documentation, adhering to CMMI level 5 processes
2. Use of UML documentation guidelines
3. Centralised repository of documents for sharing across the program
4. Well-defined induction and training programs.

4.7 Quality control

Test cases were written as soon as user stories were taken up for development (test-driven development). This improved clarity on requirements (user stories) and control over quality. Continuous testing helped to identify issues and defects at a very early stage. A test specialist was permanently assigned to the project, with the following roles and responsibilities – defining testing process & procedures, supervision of testing and approval of test results prior to customer demonstration. All unit tested modules were integrated together overnight (continuous integration) to identify build breaks. The software developed was reviewed as soon as a team member made progress in his/her user stories. The objective was to ensure that all stake holders were in the same page and the development was technically in the right direction. The users provided further feedback during sprint demos. The team conducted a retrospective meeting at the end of every sprint and devised corrective measures in the form of formal procedures, checklists and templates.

4.8 Imbibing Agile culture

Agile culture is organic – flexible and participative, encouraging cooperative social action [6, 20]. However, the ODC organisational structure and inherent India cultural dimensions [26, 27] aligned ODC team to hierarchical command and control. To enable a paradigm shift to Agile culture, the following measures were taken. All the team members were sent for training to understanding Agile values and principles. The trainings happened in multiple phases of learning, sharing experiences and improving. Scrum masters were sent for scrum master training programs and were encouraged to facilitate Agile culture. Scrum meetings (part of horizontal coordination) were made more effective, by announcing the agenda; adhering to meeting schedule, ensuring right participation and tracking all action items to closure. Team building exercises were planned to foster interpersonal relationship and informal collaboration. The exercises included, 'weekly fun at work programs', and day out program at the end of each release, among others. 'Accountability improved over time', with improved accuracy in estimating user stories, detailed metrics collection, transparency and focussed/productive meetings.

4.9 Contract type

Linear models such as waterfall have defined sequential phases, such as requirements analysis, design, construction and system/acceptance test. At the completion of these phases, payments are made. In the Agile method, the requirements evolve. There are no time-separated sequential phases, which could be taken as milestones indicating project progress and payment points. Earned value method [2] is used to track the percentage of work completed in traditional projects. In the Agile model, requirements evolved and were reprioritised prior to every sprint. Hence,

the progress could not be measured using earned value techniques, against which payment could be calculated. In the project under consideration, outsourcing contract was signed on T&M basis. The team continued till 'most of the functionality was delivered' staying within the customer's budgetary constraints.

5 Recommendations, limitations, conclusion and directions for future research

5.1 Recommendations

In Agile practice, project planning is continuous in nature. However, in the offshore-outsourced context, we recommend a detailed first phase, 'sprint zero'. A high-level design must be prepared, which includes requirements identification, overall system architecture, database design and system interfacing needs. Moreover, a proof concept program must be undertaken to test the design. This helps to determine the scope the work to be outsourced and prepare an outsourcing agreement. Large & distributed project teams are characteristic features of an offshore-outsourced business model. Contrary to Agile practice, tacit knowledge is not sufficient and formal documentation is necessary. Standard methods such as matured CMMI processes and UML documentation are suggested for achieving inter-team compatibility and smooth transition to maintenance and operations phase. A hybrid organisational structure blending both flat & hierarchical characteristics is recommended. This fosters Agile culture and also achieves higher levels of coordination that is needed in offshore-outsourced business model. The offshore vendor organisations must align feedback mechanisms and HR policies & processes to Agile culture, by achieving more transparency in performance measurement and by giving importance to the performance as a team. There must be programs for regular team building exercises, mentoring of new joiners and mentoring for continuous performance improvement. The risk profile & risk resolution techniques unique Agile offshore-outsourced projects (presented in this paper), along with the larger GDSP framework for distributed project management [11] (applied in this paper) can be used as a basis for preparing the risk management plan for Agile scrum offshore-outsourced projects, to achieve better project performance.

5.2 Limitations

Case study is a suitable research methodology for software engineering research, since it investigates contemporary phenomena in its natural context [22]. Although a representative case showing wide variation of features was selected for the case study, it may be noted that there are a multitude of features associated with software risk management that need detailed investigation. We obtained the perspective of the ODC team who are mainly responsible for managing risks at the software development level. However, we could have obtained a more holistic picture, if we had access to the customer team as well. The study was conducted according to the guidelines laid down in [22], including aspects such as case-study design, case selection, data triangulation and data analysis. The interviewers were associated with similar IT projects for a prolonged time, so that the interviewee's observations were well understood. Therefore we expect that the study would

add value to the body of knowledge in software risk management.

5.3 Conclusion

Agile projects by themselves are a distinct class of development projects. Offshore-outsourced context gives them a more unique risk profile. We investigated the risk profile of a large Agile offshore-outsourced project executed by geographically distributed teams working under two different vendors. Some of the unique innovations that we observed included the following – hybrid development model, hybrid organisational structure, techniques for improved resource utilisation and initiatives taken for transitioning the team from traditional culture to Agile. The GDSP framework [11] was found to be a good tool to identify and assess risks, from the perspective of distributed project management. The risk checklist that we developed, and corresponding risk resolution techniques observed can be used to examine the challenges in the implementation of Agile culture & practices in offshore-outsourced model. We conclude that Agile projects are good candidates for offshore-outsourced model, if risks unique to such model are identified, assessed and appropriate risk resolution techniques deployed.

5.4 Directions for future research

1. The risk profile and risk resolution techniques observed from the case study can be used as inputs for a detailed investigation (e.g., a Delphi study) to develop a framework for identifying, assessing and managing software development risks in Agile projects in the offshore-outsourced context, in line with the GDSP framework [11].
2. IT Software vendors allocate team members from a central talent pool and release them regularly to different assignments. The transient nature of resource allocation impacts (a) the knowledge level of the Agile team during the software development phase, and (b) creates a knowledge gap when transitioned to maintenance phase. We observed that, this risk was addressed through improved formal communication. However, this knowledge management area deserves detailed research.
3. Another possible area of study is, risk management in closely coupled Agile offshore-outsourced distributed projects (where the teams require higher levels and frequencies of task coordination).
4. Another area worth attention is longitudinal case studies on the influence of the risk resolution techniques on project performance, in the context of Agile offshore-outsourced projects.

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